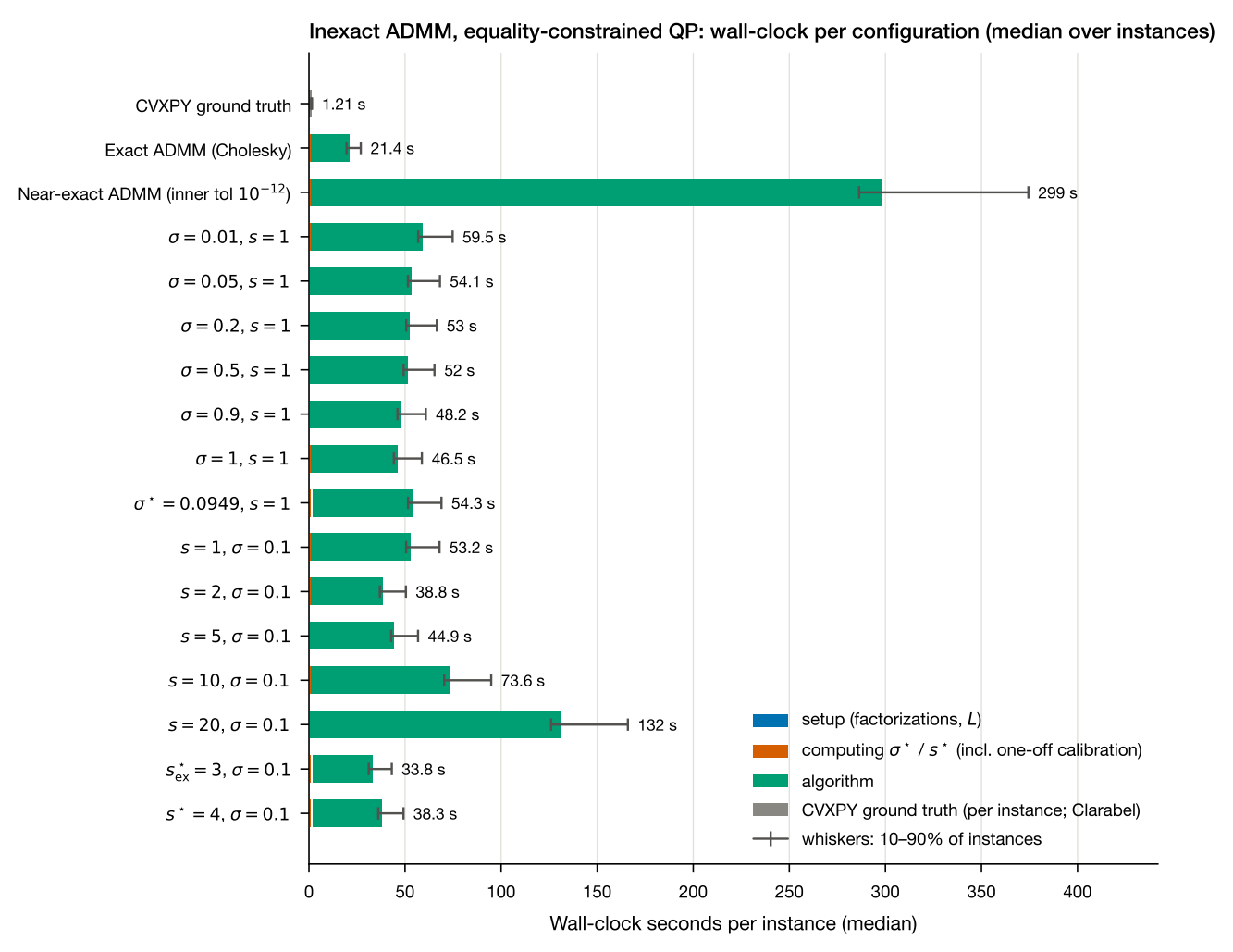
Inexact ADMM, equality-constrained QP: wall-clock per configuration (median over instances)
CVXPY ground truth
Exact ADMM (Cholesky)
Near-exact ADMM (inner tol 10^{-12})
$\sigma = 0.01, s = 1$
$\sigma = 0.05, s = 1$
$\sigma = 0.2, s = 1$
$\sigma = 0.5, s = 1$
$\sigma = 0.9, s = 1$
$\sigma = 1, s = 1$
$\sigma^\star = 0.0949, s = 1$
$s = 1, \sigma = 0.1$
$s = 2, \sigma = 0.1$
$s = 5, \sigma = 0.1$
$s = 10, \sigma = 0.1$
$s = 20, \sigma = 0.1$
$s^\star_{\text{ex}} = 3, \sigma = 0.1$
$s^\star = 4, \sigma = 0.1$
1.21 s
21.4 s
299 s
59.5 s
54.1 s
53 s
52 s
48.2 s
46.5 s
54.3 s
53.2 s
38.8 s
44.9 s
73.6 s
132 s
33.8 s
38.3 s
setup (factorizations, L)
computing $\sigma^\star$ / $s^\star$ (incl. one-off calibration)
algorithm
CVXPY ground truth (per instance; Clarabel)
whiskers: 10–90% of instances
0
50
100
150
200
250
300
350
400
Wall-clock seconds per instance (median)